Mock shell

/*SEX*
MALE=1 , FEMALE=2

TRTO1A
BP3304 = 1
PLACEBO = 0

RACE
WHITE = 1
Black or African American =2
Asian = 3
American Indian or Alaska Native =4
Native Hawaiian or Other Pacific Islander = 5
Other = 6

ETHNICITY
Hispanic or Latino = 1
Not Hispanic or Latino = 2

ALCOHOL_HISTORY
Currently Consumes =1
Previously Consumed =2
Never Consumed =3

TOBACCO_HISTORY
Currently Consumes =1
Previously Consumed =2
Never Consumed =3

*/

```
libname Table "/home/u64252598/SAS/New Folder";
run;

PROC IMPORT DATAFILE="/home/u64252598/SAS/New Folder/ADSL.xlsx"
OUT=table.ADAM
DBMS=XLSX REPLACE;
GETNAMES=YES;
QUIT;
```

```
data k;
set table.adam;
run;

proc contents data=k;
quit;

proc means data=k maxdec=0;
quit;


proc sort data=k out=adam1;
by age;
quit;

data adam4;
set adam1;
if sex="Male" then sex=1;
else if sex="Female" then sex=2;
if TRT01A="BP3304" THEN TRT01A=1;
ELSE IF TRT01A="Placebo" THEN TRT01A=0;
IF RACE="Asian" THEN RACE=3;
IF RACE="White" THEN RACE=1;
IF RACE="American Indian or Alaska Native" THEN RACE=4;
IF RACE="Native Hawaiian or Other Pacific Islander" THEN RACE=5;
IF RACE="Black or African American" THEN RACE=2;
IF RACE="Other" THEN RACE=6;
IF ETHNICITY = "Hispanic or Latino" THEN ETHNICITY=1;
IF ETHNICITY = "Not Hispanic or Latino" THEN ETHNICITY=2;
IF ALCOHOL_HISTORY = "Currently Consumes" THEN ALCOHOL_HISTORY
=1;
IF ALCOHOL_HISTORY = "Previously Consumed" THEN
ALCOHOL_HISTORY =2;
IF ALCOHOL_HISTORY ="Never Consumed" THEN ALCOHOL_HISTORY =3;
IF TOBACCO_HISTORY ="Currently Consumes" THEN
TOBACCO_HISTORY=1;
IF TOBACCO_HISTORY ="Previously Consumed" THEN
TOBACCO_HISTORY=2;
IF TOBACCO_HISTORY = "Never Consumed" THEN TOBACCO_HISTORY=3;
run;
```

```
data adam5;
set adam4;output;
TRT01A=2;output;
run;

proc sort data=adam5 out=adam6;by TRT01A;
quit;

proc summary data=adam6 ;
by trt01a;
var age;
output out=age_1 n=_n_ mean=_mean_ std=_std_ median=_median_
min=_min_ max=_max_;
quit;

DATA age_2;
set age_1;
Meansd=put(_mean_,5.1)||"("||put(_std_,6.2)||")";
minmax=put(_min_,4.0)||","||put(_max_,4.0);
n=put(_N_,4.0);
Median=put(_median_,5.1);
drop _:;
run;

proc transpose data=age_2 out=age_3 prefix=tret;
var n Meansd median minmax;
id trt01a;
quit;

data age_4;
length newvar $ 40;
set age_3;
if _Name_="n" then newvar="        N";
else if _Name_="Meansd" then newvar="       Mean(SD)";
else if _Name_="Median" then newvar="       Median";
else if _Name_="minmax" then newvar="       Min,Max";
drop _Name_;
Suborder=1;
run;
```

```
data dummy;
length newvar $ 40;
newvar="Age(Years)";
Suborder=0;
run;

data age;
set dummy age_4 ;
order=1;
run;
```

/* Gender statistics */

```
proc freq data=adam6 noprint;
by trt01a;
table sex/out=gen_1;
run;

data gen_2;
set gen_1;
np=put(count,4.0)||"("||put(Percent,4.1)||")";
run;

proc sort data=gen_2;
by sex;
quit;

proc transpose data=gen_2 out=gen_3 (drop=_NAME_) prefix=tret;
var np;
id trt01a;
by sex;
run;

data gen_4;
length newvar $ 60;
set gen_3;
if sex=1 then newvar="          Male";
else if sex=2 then newvar="          Female";
Suborder=2;
run;
```

```
data dummy1;
length newvar $ 60;
newvar="Gender [n (%)]a";
Suborder=0;
run;


data gender;
set dummy1 gen_4;
order=2;
run;
```

/* Ethnicity */

```
proc freq data=adam6 noprint;
by trt01a;
table ETHNICITY/out=eth_1;
run;

data eth_2;
set eth_1;
np=put(count,4.0)||"("||put(Percent,4.1)||")";
run;

proc sort data=eth_2;
by ETHNICITY;
quit;

proc transpose data=eth_2 out=eth_3 (drop=_NAME_) prefix=tret;
var np;
id trt01a;
by ETHNICITY;
run;

data eth_4;
length newvar $ 60;
set eth_3;
if ETHNICITY=1 then newvar="          Hispanic or Latino";
else if ETHNICITY=2 then newvar="          Not Hispanic or Latino";
Suborder=3;
run;
```

```
data dummy2;
length newvar $ 60;
newvar="Ethnicity [n (%)]a";
Suborder=0;
run;


data Ethnicity;
set dummy2 eth_4;
order=3;
run;


/* Race */

proc freq data=adam6 noprint;
by trt01a;
table RACE/out=race_1;
run;

data race_2;
set race_1;
np=put(count,4.0)||"("||put(Percent,4.1)||")";
run;

proc sort data=race_2;
by RACE;
quit;

proc transpose data=race_2 out=race_3 (drop=_NAME_) prefix=tret;
var np;
id trt01a;
by RACE;
run;

data race_4;
length newvar $ 90;
set race_3;
if RACE=1 then newvar="          White";
else if RACE=2 then newvar="          Black or African American";
```

```
else if RACE=3 then newvar="        Asian";
else if RACE=4 then newvar="        American Indian or Alaskan Native";
else if RACE=5 then newvar="        Native Hawaiian or Other Pacific Islander";
else if RACE=6 then newvar="        Other";
Suborder=4;
run;

data dummy3;
length newvar $ 90;
newvar="Race [n (%)]a";
Suborder=0;
run;


data RACE;
set dummy3 race_4;
order=4;
run;

data final;
set age gender Ethnicity RACE;
keep newvar tret0 tret1 tret2 order Suborder;
run;

ods pdf file="/home/u64252598/SAS/New Folder/table_out2.pdf";

proc report data=final nowd headline headskip split="*"
style(report)=[rules=groups frame=hsides]
style(header)=[borderbottomwidth=1]
style(column)=[bordertopwidth=0 borderbottomwidth=0];

column(order Suborder newvar tret1 tret0 tret2);

define order/group noprint;
define Suborder/group noprint ;

define newvar /"" width=40
style=[just=left];

define tret1/"BP3304*(N=31)";
define tret0/"Placebo*(N=29)";
```

```
define tret2/"Overall*(N=60)";

break after order/skip;

compute before _page_;
line "";
line @9 "14.1.2.1 Subject Demographics and Baseline Characteristics";
line @19 "Safety Population";
line " ";
endcomp;

compute after;
line @4 "Reference: Listing 16.2.4.1";
line @5 "Percentages are based on the number of subjects in the population.";
line @4 "Note: SD = standard deviation, Min = Minimum, Max = Maximum";
line "";
endcomp;
run;

ods pdf close;
```

/* 14.1.2.1 Subject Demographics and Baseline Characteristics */
/* Safety Population */

/* Height_cm */

```
PROC IMPORT DATAFILE="/home/u64252598/SAS/New Folder/ADSL.xlsx"
OUT=ADAM01
DBMS=XLSX REPLACE;
GETNAMES=YES;
QUIT;

data adam02;
set adam5 (keep=height_cm weight_kg bmi TRT01A);
run;


proc sort data=adam02 out=adam03;by TRT01A;
quit;
```

```
proc summary data=adam03;
var height_cm;
by TRT01A;
output out=hgt_1 n=_n_ mean=_mean_ stddev=_std_ median=_median_
min=_min_ max=_max_;
quit;


data hgt_2;
set hgt_1;
meansd=put(_mean_,4.1)||"("||put(_std_,5.2)||")";
median=put(_median_,4.1);
minmax=put(_min_,4.0)||","||put(_max_,4.0);
n=put(_n_,4.0);
drop _:;
run;

PRoc transpose data=hgt_2 out=hgt_3 prefix=tret;
var n meansd median minmax;
id TRT01A;
quit;


data hgt_4;
length newvar $ 60;
set hgt_3;
if _NAME_ = "n" then newvar="       N";
else if _NAME_="meansd" then newvar="       Mean(SD)";
else if _NAME_="median" then newvar="       Median";
else if _NAME_="minmax" then newvar="       Min,Max";
drop _:;
suborder=1;
run;

data dummy01;
length newvar $ 60;
newvar="Height (cm)";
suborder=0;
run;
```

```
data Height;
set dummy01 hgt_4;
order=1;
run;


/* Macro */

%macro proj(var=,dset1=
,dset2=,dset3=,dset4=,newvar=,dummy=,final=,order1=,subordern0=,subordern
1=);

PROC IMPORT DATAFILE="/home/u64252598/SAS/New Folder/ADSL.xlsx"
OUT=ADAM01
DBMS=XLSX REPLACE;
GETNAMES=YES;
QUIT;

data adam4;
set ADAM01;
if sex="Male" then sex=1;
else if sex="Female" then sex=2;
if TRT01A="BP3304" THEN TRT01A=1;
ELSE IF TRT01A="Placebo" THEN TRT01A=0;
IF RACE="Asian" THEN RACE=3;
IF RACE="White" THEN RACE=1;
IF RACE="American Indian or Alaska Native" THEN RACE=4;
IF RACE="Native Hawaiian or Other Pacific Islander" THEN RACE=5;
IF RACE="Black or African American" THEN RACE=2;
IF RACE="Other" THEN RACE=6;
IF ETHNICITY = "Hispanic or Latino" THEN ETHNICITY=1;
IF ETHNICITY = "Not Hispanic or Latino" THEN ETHNICITY=2;
IF ALCOHOL_HISTORY = "Currently Consumes" THEN ALCOHOL_HISTORY
=1;
IF ALCOHOL_HISTORY = "Previously Consumed" THEN
ALCOHOL_HISTORY =2;
IF ALCOHOL_HISTORY ="Never Consumed" THEN ALCOHOL_HISTORY =3;
IF TOBACCO_HISTORY ="Currently Consumes" THEN
TOBACCO_HISTORY=1;
IF TOBACCO_HISTORY ="Previously Consumed" THEN
TOBACCO_HISTORY=2;
```

```
IF TOBACCO_HISTORY = "Never Consumed" THEN TOBACCO_HISTORY=3;
run;

data adam5;
set adam4;output;
TRT01A=2;output;
run;

data adam02;
set adam5 (keep=height_cm weight_kg bmi TRT01A);
run;


proc sort data=adam02 out=adam03;
by TRT01A;
quit;

proc summary data=adam03;
var &var;
by TRT01A;
output out=&dset1 n=_n_ mean=_mean_ stddev=_std_ median=_median_
min=_min_ max=_max_;
quit;


data &dset2;
set &dset1;
meansd=put(_mean_,4.1)||"("||put(_std_,5.2)||")";
median=put(_median_,4.1);
minmax=put(_min_,4.0)||","||put(_max_,4.0);
n=put(_n_,4.0);
drop _:;
run;

PRoc transpose data=&dset2 out=&dset3 prefix=tret;
var n meansd median minmax;
id TRT01A;
quit;


data &dset4;
```

```
length newvar $ 60;
set &dset3;
if _NAME_ = "n" then newvar="       N";
else if _NAME_="meansd" then newvar="       Mean(SD)";
else if _NAME_="median" then newvar="       Median";
else if _NAME_="minmax" then newvar="       Min,Max";
drop _:;
suborder=&subordern1;
run;

data &dummy;
length newvar $ 60;
newvar=&newvar;
suborder=&subordern0;
run;

data &final;
set &dummy &dset4;
order=&order1;
run;

%mend;

/* height_cm weight_kg bmi TRT01A */

/* Weight */
%proj(var=weight_kg,dset1=wtg_1
,dset2=wtg_2,dset3=wtg_3,dset4=wtg_4,newvar="Weight
(kg)",dummy=dummy02,final=weight,order1=2,subordern0=0,subordern1=2);

/* Body mass index */
%proj(var=BMI,dset1=bmi_1
,dset2=bmi_2,dset3=bmi_3,dset4=bmi_4,newvar="Body Mass Index
(kg/m2)",dummy=dummy03,final=BMI,order1=3,subordern0=0,subordern1=3);

data final2;
set Height weight BMI;
run;

/* Report  */
```

```
proc report data=final2 nowd headline headskip split="*";

column(order suborder newvar tret1 tret0 tret2);

define order / group noprint;
define suborder / group noprint;
define newvar / display "";

define tret1/"BP3304*(N=45)";
define tret0/"Placebo*(N=35)";
define tret2/"Overall*(N=80)";

break after order/skip;

compute before _page_;
line "";
line @9 "14.1.2.1 Subject Demographics and Baseline Characteristics";
line @19 "Safety Population";
line " ";
endcomp;

compute after;
line @4 "Reference: Listing 16.2.4.1";
line @4 "Note: SD = standard deviation, Min = Minimum, Max = Maximum";
line "";
endcomp;

run;
```

OUTPUT OF TABLE1

14.1.2.1 Subject Demographics and Baseline Characteristics
Safety Population

	BP3304 (N=31)	Placebo (N=29)	Overall (N=60)
Age(Years)			
N	45	35	80
Mean(SD)	53.1(12.66)	51.0(10.34)	52.2(11.68)
Median	52.0	48.0	49.5
Min,Max	35, 73	36, 70	35, 73
Gender [n (%)]a			
Male	20(44.4)	14(40.0)	34(42.5)
Female	25(55.6)	21(60.0)	46(57.5)
Ethnicity [n (%)]a			
Hispanic or Latino	16(35.6)	7(20.0)	23(28.8)
Not Hispanic or Latino	29(64.4)	28(80.0)	57(71.3)
Race [n (%)]a			
White	9(20.0)	9(25.7)	18(22.5)
Black or African American	8(17.8)	6(17.1)	14(17.5)
Asian	4(8.9)	5(14.3)	9(11.3)
American Indian or Alaskan Nat	10(22.2)	4(11.4)	14(17.5)
Native Hawaiian or Other Pacif	6(13.3)	4(11.4)	10(12.5)
Other	8(17.8)	7(20.0)	15(18.8)

Reference: Listing 16.2.4.1
Percentages are based on the number of subjects in the population.
Note: SD = standard deviation, Min = Minimum, Max = Maximum

OUTPUT OF TABLE2

14.1.2.1 Subject Demographics and Baseline Characteristics
Safety Population

	BP3304 (N=45)	Placebo (N=35)	Overall (N=80)
Height (cm)			
N	45	35	80
Mean(SD)	164(9.42)	166(10.64)	165(9.93)
Median	163	168	165
Min,Max	146, 184	144, 195	144, 195
Weight (kg)			
N	45	35	80
Mean(SD)	65.5(8.46)	69.8(13.09)	67.4(10.88)
Median	66.1	68.8	66.4
Min,Max	47, 84	43, 95	43, 95
Body Mass Index (kg/m2)			
N	45	35	80
Mean(SD)	24.5(3.96)	25.8(6.17)	25.1(5.06)
Median	24.3	25.8	24.8
Min,Max	15, 33	14, 42	14, 42

Reference: Listing 16.2.4.1
Note: SD = standard deviation, Min = Minimum, Max = Maximum